Nonequilibrium training dynamics and scaling laws of language models

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9 November 2026

ABSTRACT
Large language models (LLMs) exhibit empirical scaling laws, yet the mechanistic origin of these power laws remains poorly understood. We propose a stochastic multiscale theory of LLM training dynamics, grounded in a formal analogy with nonequilibrium systems such as turbulence and cosmic structure formation, where scaling laws emerge naturally from scale-space transport processes. Central to the theory is a scale-dependent loss $L_r(r,s)$, the held-out loss when usable context is restricted to distance r at training step s. Its negative gradient $E_X(r,s) \equiv -\partial L_r/\partial r$ defines a *loss spectrum* in direct analogy with the turbulence energy spectrum. Training dynamics is modeled as a biased random walk with scale-dependent waiting time $\tau_r \propto r^{-\gamma}$ in r-space, governed by a Fokker-Planck equation whose exact solution yields an analytically tractable loss spectrum with inertial-range scaling $E_X \propto r^{-(1+\gamma)}$ and an exponential cutoff at an effective scale $r = \xi$. The key exponent $\gamma = \alpha_H/3 = 1/6$ is determined self-consistently from Heaps' law ($\alpha_H \approx 1/2$). Decomposing the total loss into $L_\infty(s)$ (the residual full-context loss, characterizing the *breadth* of the learning frontier) and $\Delta L_r(s)$ (the *depth* of context dependence), we identify a two-phase training dynamics that bifurcates at a critical step s^* into two simultaneous modes: **Mode A** (downscale compression), with a shrinking effective scale $\xi_A \propto s^{-3/5}$ and growing $\Delta L_r \propto s^{+1/10}$; and **Mode B** (upscale frontier absorption), with an expanding scale $\xi_B \propto s^{+3/5}$ and decreasing $L_\infty \propto s^{-1/10}$. The theory directly predicts the empirical loss-vs-data scaling exponent $\alpha_D = \gamma/(2-2\gamma) = 1/10$, so that $L \propto D^{-\alpha_D}$. Treating model size N as the effective Langevin time yields $\alpha_N = \lambda\gamma/(2-2\gamma)$, where the asymmetry parameter $\lambda \leqslant 1$ encodes the fraction of parameters contributing to learning; $\lambda = 3/4$ gives $\alpha_N = 3/40 \approx 0.075$, consistent with the empirical Kaplan value. The full scaling form $L(N,D) = [(N_c/N)^\lambda + D_c/D]^{\alpha_D}$ is recovered from the harmonic combination of two independent bottlenecks. The framework yields falsifiable predictions, including a universal loss spectrum $E_X \propto r^{-7/6}$, and provides diagnostics for long-context generalization, data selection, training curriculum, and architecture design.

Key words: Language modeling; Scaling laws; Nonequilibrium; Turbulence; Cosmic structure; Mass/energy cascade

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1 INTRODUCTION

Large transformer-based language models obey strikingly empirical scaling laws: held-out cross-entropy falls as a power law in parameter count N, dataset size D, and training compute C across many orders of magnitude [1, 2, 3]. These laws now guide practical compute allocation [3], yet their mechanistic origin remains unclear: A natural place to look is the structure of predictive information across *context distance*. Self-attention couples every token to every other within the context window [4], yet long-context performance is uneven and sensitive to where information sits in the input [5], and dedicated long-range memory mechanisms measurably reduce perplexity [6, 7]. At the data level, natural language shows a power-law decay of mutual information with token separation [8], with different linguistic relationships preferentially encoded at different scales [9]. This structure has recently been used to predict data-limited scaling exponents directly from conditional-entropy decay rates [10], while mutual-information scaling laws have been proposed as the key quantity governing long-context capability [11]. On the dynamic side, training exhibits at least one abrupt transition marking the sudden emergence of long-range in-context prediction, coincident with the formation of induction heads [12]. Together these results show that both the static (data) and dynamic (training) sides of context-distance structure shape power-law scaling behavior, yet no existing account ties the two together quantitatively.

Power-law scaling is, more broadly, the hallmark of driven, multiscale nonequilibrium systems, which share four ingredients: *openness* (an external drive holding the system away from equilibrium), *nonlinear interactions* (coupling degrees of freedom across scales), *fluctuations* (seeding structural transitions), and *dissipation* (stabilizing the resulting structure). Two canonical physical examples make the pattern concrete, inspiring the language modeling

Turbulence in a stirred cup of coffee is a canonical case: energy injected at the stirring scale L (*openness*) cascades through a hierarchy of eddies coupled by the Navier-Stokes advection term (*nonlinear interactions*), seeded by stochastic velocity perturbations that

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trigger eddy breakup (*fluctuations*), down to the Kolmogorov scale η where viscosity dissipates the accumulated energy (*dissipation*). This can be captured evocatively by Richardson's verse [13]:

"Big whirls have little whirls, that feed on their velocity;
and little whirls have lesser whirls, and so on to viscosity."

Under Kolmogorov's (K41) self-similarity assumption, the inertial-range energy spectrum (distribution of energy across wavenumber k) is fixed by the energy flux ε alone, giving the $-5/3$ law [14]; the Pao spectrum extends this empirically into the dissipation range [15]:

$$E(k) = C_K \varepsilon^{2/3} k^{-5/3} \exp\left[-\tfrac{3}{2} C_K (\eta k)^{4/3}\right], \tag{1}$$

with $C_K \approx 1.5$–1.7 a near-universal constant.Despite the complexity of the underlying Navier–Stokes dynamics, the energy spectrum collapses onto one universal curve, exhibiting a power law at large scales and an exponential cutoff at the small dissipation scale η.

Gravitational clustering of dark matter shows the same phenomenology in reverse. In the excursion-set (Press–Schechter) picture, halo formation is the first barrier crossing of a stochastic trajectory in smoothed density space [16, 17, 18]. The same four ingredients apply: continuous gravitational condensation of free particles into bound structures supplies *openness*; nonlinear gravitational infall coupling density fluctuations across scales supplies the *nonlinear interactions*; the primordial stochastic density field seeds the *fluctuations* forming primordial halo structures; and virialization, which locks mass into stable haloes, plays the role of *dissipation*. For hierarchical structure formation, small haloes form first and merge into larger ones, an *inverse* mass cascade across halo mass scales [19, 20], mirroring turbulence's *direct* energy cascade:

"Little haloes have big haloes, that feed on their mass;
and big haloes have greater haloes, and so on to growth."

The resulting halo mass function, or mass spectrum for distribution of mass across mass scales m_h, takes the same qualitative form as Eq. (1): a power law at small mass with an exponential cutoff at a characteristic scale M^*:

$$f_{\rm PS}(m_h) = \frac{3+n}{3\sqrt{2\pi}} \frac{1}{M^*} \left(\frac{m_h}{M^*}\right)^{-\frac{3-n}{6}} \exp\left[-\frac{1}{2}\left(\frac{m_h}{M^*}\right)^{\frac{3+n}{3}}\right]. \tag{2}$$

Turbulence and gravitational clustering thus sit at opposite ends of the same cascade phenomenon — energy flowing down in scale versus mass flowing up — yet both follow the same openness/nonlinearity/fluctuation/dissipation logic within a common stochastic framework, and both yield a power law truncated by a single characteristic scale (Appendices A–B).

Language modeling exhibits the same qualitative signature of a driven, nonlinear, multiscale system. Token and word statistics are heavy-tailed (Zipf/Heaps laws) [21, 22], and next-token prediction draws simultaneously on local syntax and long-range context structure—a hierarchy of interacting scales directly analogous to eddies in turbulence or haloes in cosmic structure formation. Mapped onto the four nonequilibrium ingredients: the continuously fed data stream supplies *openness*, injecting unresolved predictive challenges at progressively larger context scales and driving the training dynamics away from any equilibrium; self-attention and MLP nonlinearities provide *nonlinear coupling* that redistributes learning across token positions and representational scales; Stochastic training dynamics including minibatch sampling noise plausibly supply the *fluctuations* that continuously seed the formation of new predictive structures at growing scales; and the consolidation of stable predictive structures at the learning frontier—locking in new capabilities—provides the *dissipative* channel that stabilizes the resulting representations. This correspondence motivates a *scale-space* view of LLM training: context utilization is not fixed, but progressively grows across different scales over training, with an effective dependency length that expands as training proceeds. Just as turbulence distributes kinetic energy across eddy scales and gravitational clustering distributes mass across halo scales, training distributes learned capabilities and unresolved loss across context scales r—giving rise to a *loss spectrum* $E_X(r,s)$ that is the central object of this paper.

This paper develops a nonequilibrium multiscale theory of LLM training dynamics by introducing a context scale r that indexes dependency distance in natural language and modeling training as a biased random walk in r-space with scale-dependent waiting time $\tau_r \propto r^{-\gamma}$. The associated Fokker–Planck equation yields a closed-form loss spectrum, structurally analogous to the turbulence energy spectrum and halo mass function (Eqs. (1)–(2) derived from the same nonequilibrium framework in Appendices A–B), that provides a mechanistic account of empirical scaling laws with falsifiable, checkpoint-testable predictions.

2 BACKGROUND AND DEFINITIONS

2.1 Language Modeling Loss

Given a token sequence $x_{1:T}$, a causal language model with parameters θ at optimizer step s defines the next-token distribution

$$p_{\theta(s)}(x_{1:T}) = \prod_{t=1}^{T} p_{\theta(s)}(x_t \mid x_{<t}), \tag{3}$$

with per-token negative log-likelihood

$$\ell_t(\theta(s)) = -\log p_{\theta(s)}(x_t \mid x_{<t}). \tag{4}$$

The held-out loss is the expected cross-entropy

$$L(\theta(s); N, D) = \mathbb{E}_{(x_{<t}, x_t)\sim \text{data}}\left[\ell_t(\theta(s))\right], \tag{5}$$

where N is the parameter count and D the training set size. Empirical scaling describe the near-optimally trained held-out loss as a function of N, D, and compute C [1, 3], with reported scaling $L \propto N^{-0.076}$, $L \propto D^{-0.095}$, $L \propto C^{-0.05}$ (N for non-embedding parameters).

2.2 Context Scale and the Loss Spectrum

Natural language carries statistical structures across many scales, from adjacent-token correlations to document-level dependencies. The scalar loss L in Eq. (5) aggregates all of these scales into a single number, obscuring how information is distributed across scales. To resolve this, we introduce a context-scale variable r (measured in tokens) as the maximum dependency distance the model is permitted to use when predicting the next token.

Restricting the model's receptive field to the most recent r tokens defines a scale-dependent predictive distribution,

$$p^{(r)}_{\theta(s)}(x_t \mid x_{<t}) \equiv p_{\theta(s)}\big(x_t \mid x_{\max(1,t-r):t-1}\big), \tag{6}$$

and a corresponding scale-dependent held-out loss $L_r(r,s)$, obtained by evaluating Eq. (5) with $p^{(r)}_{\theta(s)}$ in place of $p_{\theta(s)}$. By construction, L_r is non-increasing in r:

$$L_r(1,s) \geqslant L_r(r,s) \geqslant L_r(r_{\max}, s) \equiv L_\infty(s),$$

where $r_{\max}$ is the model's context window and $L_\infty(s)$ is the full-context loss remaining after all available context has been used at step s. We also define a context-dependent loss $\Delta L_r(s) \equiv L_r(1,s) - L_\infty(s)$ as the total loss attributable to context beyond a single token.

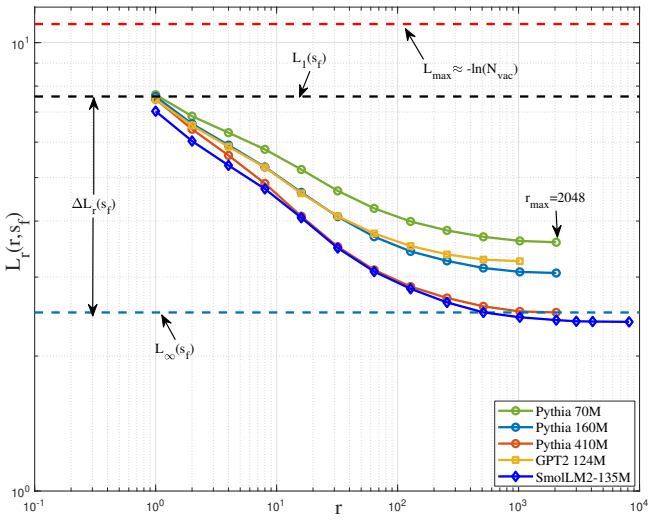


Figure 1. Scale-dependent loss $L_r(r,s)$ as a function of context scale r at the final training step $s = s_f$ for several language models of different sizes. The maximum loss $L_{\max} = \ln(N_{voc})$, determined by the vocabulary size $N_{\rm voc}$, is approached as $r \to 1$. The full-context loss $L_\infty(s)$ is attained when the full context window $r_{\max}$ is used. The total context-dependent loss $\Delta L_r(s) = L_r(1,s) - L_\infty(s)$ is distributed across context scale r. Larger models achieve both a smaller $L_\infty(s)$ and a larger total $\Delta L_r(s)$, indicating more efficient utilization of context at all scales.

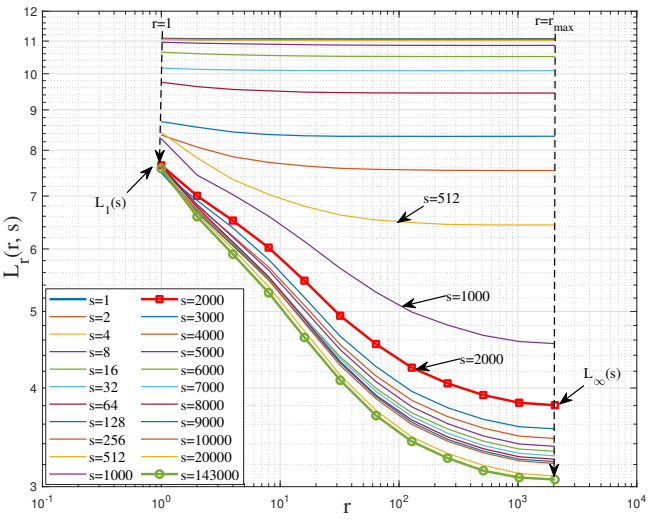


Figure 2. Loss $L_r(r,s)$ as a function of context scale r at different training steps s for the Pythia-160M model. Two distinct phases of training dynamics are visible. **Phase I** ($s < s^* \approx 2000$): the loss decreases rapidly across all scales, with large-scale losses dropping faster than small-scale losses, indicating that the model progressively extends its predictive reach to longer context distances (upscale cascade). **Phase II** ($s > s^*$): the single-token loss $L_1(s)$ has largely converged, and further improvement concentrates at shorter context scales as the model refines local predictions (downscale cascade).

We define the *loss spectrum* as the (negative) rate at which loss is recovered as the usable context scale grows,

$$E_X(r,s) \equiv -\frac{\partial L_r(r,s)}{\partial r} \geqslant 0, \qquad P_X(r,s) \equiv \frac{E_X(r,s)}{\Delta L_r(s)}. \tag{7}$$

Integrating Eq. (7) from a given scale r to $r_{\max} \to \infty$ yields

$$L_r(r,s) = L_\infty(s) + \int_r^\infty E_X(r',s)\, dr', \tag{8}$$

and, evaluating at $r = 1$,

$$\Delta L_r(s) = \int_1^\infty E_X(r',s)\, dr'. \tag{9}$$

By construction, $P_X(r,s)$ is the normalized spectrum, i.e., the fraction of the total context-dependent loss reduction attributable to scale r. The quantity $E_X(r,s)$ plays the role of an energy spectrum $E(k)$ in turbulence or mass function of haloes, and $\Delta L_r(s)$ for the total "energy" or "mass" distributed across scale r.

Training dynamics thus reduces to tracking how the model learned capability to exploit multiscale context ($\Delta L_r(s)$) and residual full-context loss ($L_\infty(s)$) jointly evolve across all context scales r. The central question of this work is: *what governs this multiscale evolution, and what scaling laws does it imply?*

To make this concrete, we now describe how $L_r(r,s)$ and its gradient $E_X(r,s)$ are estimated from model checkpoints at fixed training step s. For a fixed model checkpoint at step s, we estimate $L_r(r,s)$ and $E_X(r,s)$ as follows (Figs. 1 and 2):

(i) Select a held-out token stream (e.g. the WikiText-103).
(ii) Compute the reference loss $L_r(r_{\max},s) \equiv L_\infty(s)$ under standard causal evaluation.
(iii) For each r ($r \in \{1,2,4,8,\ldots,r_{\max}\}$), compute the held-out loss with attention restricted to the trailing r tokens (Eq. (6)), truncating context and re-indexing positions consistently.
(iv) Estimate $E_X(r,s)$ as a finite difference of $L_r(r,s)$ across adjacent grid points (Eq. (7)).

The context window $r_{\max}$ is architecture-dependent: 1024 tokens for GPT-2, and 2048 tokens for the Pythia model suite evaluated here.

2.3 Two Fundamental Loss Quantities and Learning Modes

Our scale-resolved framework identifies two fundamental loss quantities with distinct physical meanings, each governed by a corresponding learning mode:

(i) $\Delta L_r(s)$*: Depth of context dependence (Mode A).* This quantity measures the total predictive value of multi-scale context—how much better the model predicts when given full context versus only a single preceding token. It quantifies the degree to which the model has learned to *rely on* contextual structure beyond the immediately adjacent token. $\Delta L_r(s)$ increases monotonically with training (Fig. 3) as the model develops increasingly sophisticated representations in which meaning is distributed across multiple context scales, making single-token predictions inherently harder relative to full-context predictions. In Phase II, $\Delta L_r(s) \propto s^{+1/10}$. A growing ΔL_r is not a sign of deterioration but of deepening understanding: it reflects the model's increasing capacity to extract predictive value from contextual dependencies.

(ii) $L_\infty(s)$*: Residual full-context loss (Mode B).* This is the held-out loss evaluated with the full context window available—the quantity measured in most existing scaling studies. It reflects the model's remaining inability to exploit statistical dependencies across *all* context scales: not because context is unavailable, but because the model has not yet learned to use full-context effectively at the learning frontier. $L_\infty(s)$ decreases monotonically with training as the learning frontier advances outward and the model progressively masters longer-range dependencies. In Phase II, $L_\infty(s) \propto s^{-1/10}$ (Fig. 3).

These two loss quantities correspond to two complementary learning modes that operate simultaneously:

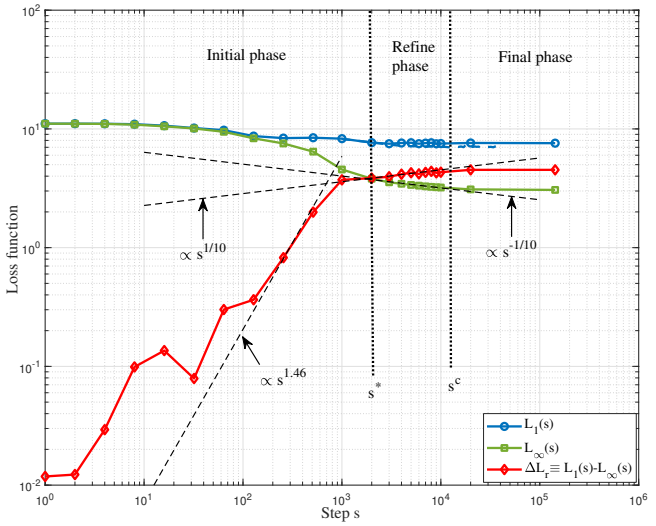


Figure 3. Evolution of fundamental loss quantities with training step s for the Pythia-160M model. The single-token loss $L_1(s)$ (blue curve) decreases monotonically and saturates earliest, around $s \approx s^*$. The full-context loss $L_\infty(s)$ (green curve) initially tracks $L_1(s)$, then follows scaling $L_\infty \propto s^{-1/10}$ in phase II after s^* (defined as $\Delta L_r(s^*) = L_\infty(s^*)$), and saturates much later in the final phase. The context-dependent loss $\Delta L_r(s)$ (red curve) also exhibits three distinct regimes: a rapid initial rise during Phase I (the plotted points rise as $s^{0.9}$–$s^{1.1}$ for $s = 16$–1000, steepest local slope 1.27; the dashed line is the predicted $s^{1.46}$ of Eq. (46)), a shallower power-law growth $\Delta L_r(s) \propto s^{1/10}$ during Phase II, and gradual saturation in final phase. These scaling regimes are the primary focus of the theoretical framework developed in this work.

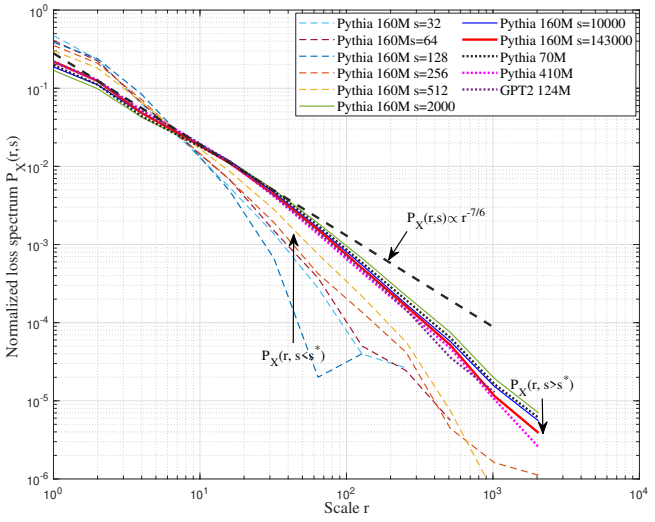


Figure 4. Normalized loss spectrum $P_X(r,s)$ as a function of context scale r at different training steps s for the Pythia-160M model. During Phase I ($s < s^*$), the fraction of loss attributed to large scales increases over time while that at small scales decreases, consistent with an upscale cascade as the model expands its predictive reach. During Phase II ($s > s^*$), the spectrum shifts back toward smaller r: large-scale contributions diminish as the model refines and consolidates learned information into short-range representations, consistent with a downscale (inverse) cascade. In Phase 2, the spectrum converges toward the theoretically predicted power-law tail $P_X(r,s) \propto r^{-7/6}$, shown as a dashed reference line.

Mode A (depth, downscale compression): The model deepens its understanding of multi-scale context structure by compressing large-scale representations into efficient short-range encodings and increasing ΔL_r.

Mode B (breadth, upscale frontier expansion): The model expands its learning frontier by absorbing new long-range predictive structure from the data and decreasing L_∞.

Together, $L_\infty(s)$ and $\Delta L_r(s)$ fully characterize the two complementary aspects of the model's learning progress: how well it predicts at full context (breadth), and how much it benefits from the multiscale context structure (depth). The present work develops a unified stochastic theory for both learning modes and derives the phenomenological scaling laws that govern their evolution.

2.4 Training Dynamics: Two-phase Cascade

Tracking $L_r(r,s)$, $E_X(r,s)$, $\Delta L_r(s)$ and $L_\infty(s)$ across checkpoints (Figs. 1, 2, 3, 4) reveals a clear picture of distinct phases on how model capability develops over training via two modes of learning.

(i) **Phase I (upscale cascade), $s < s^*$.** This is the initial phase before the training dynamics reach a statistically steady state. The loss $L_r(r,s)$ falls rapidly across all scales, with large-scale losses dropping faster than small-scale ones. Predictive structure captured first at short range progressively extends to longer context distances, reflected by a growing effective context scale $\xi(s)$ (exponential cutoff scale in Eq. (14)). The context-dependent loss $\Delta L_r(s)$ increases steeply ($\propto s^{1.46}$) while the full-context loss $L_\infty(s)$ decreases sharply. This constitutes an *upscale cascade*: the model rapidly expands the breadth of its learning before consolidating depth.

(ii) **Phase II (Bifurcation: downscale cascade for Mode A and upscale cascade for Mode B), $s^* < s < s^c$.** After the critical step s^*, defined by the equipartition condition $\Delta L_r(s^*) = L_\infty(s^*)$, the training dynamics reaches a statistically steady state and bifurcates into two simultaneous learning modes:

Mode A (downscale cascade, depth): The context-dependent loss grows as $\Delta L_r(s) \propto s^{+1/10}$ (Fig. 3), while the normalized spectrum $P_X(r,s)$ shifts toward shorter scales, converging to $P_X(r,s) \propto r^{-7/6}$ (Fig. 4). The model deepens its context dependence by compressing large-scale representations into efficient short-range encodings, with a shrinking effective context scale $\xi_A \propto s^{-3/5}$, a *downscale (direct) cascade* of learned structure from large to small scales.

Mode B (upscale frontier, breadth): The full-context loss decreases as $L_\infty(s) \propto s^{-1/10}$ (Fig. 3), reflecting the model's continued absorption of new long-range predictive structure as the learning frontier advances outward, with a growing effective scale $\xi_B \propto s^{+3/5}$—an *upscale (inverse) cascade* extending the model's effective reach to progressively larger context distances.

The two modes share the same scaling exponent $1/10$ but operate on complementary components of the loss and evolve in opposite directions in time. The theory is presented in Section 3.1.

(iii) **Phase III (saturation), $s > s^c$.** Both $\Delta L_r(s)$ and L_∞ depart from the power law and saturate, and the marginal loss reduction per training step diminishes toward zero.

This picture is formalized as an information cascade across context-distance scales, in direct analogy with the energy cascade of turbulence. Phase I constitutes an upscale transfer of predictive structure from short to long range. Phase II bifurcates into two simultaneous processes: a downscale cascade that compresses and refines existing large-scale structure into efficient local representations (Mode A), and a continued upscale cascade that absorbs new long-range structure at the advancing learning frontier (Mode B).

3 NONEQUILIBRIUM TRAINING DYNAMICS

3.1 Random Walk in r-Space for Context Learning

We model the evolution of training dynamics by interpreting the normalized loss spectrum $P_X(r,s)$ as the probability distribution of *capability carriers* across context-scale space. A capability carrier at scale r represents a unit of learned predictive capability at context distance r—the model's acquired ability to exploit statistical dependencies at that scale, contributing $E_X(r,s)\,dr = \Delta L_r(s)\cdot P_X(r,s)\,dr$ to the total learned predictive capability $\Delta L_r(s)$. The loss spectrum $E_X(r,s)$ therefore measures the distribution of learned capability across scales, in direct analogy with the turbulence energy spectrum $E(k)$ measuring the distribution of kinetic energy across k-space.

At the onset of training ($s=0$), the model has no learned predictive capability at any scale: $E_X(r,0)=0$ everywhere and $\Delta L_r(0)=0$, meaning the model predicts equally poorly regardless of how much context is provided. As training begins, capability is first created at short scales ($r\sim 1$), where high-frequency Zipfian patterns provide strong, consistent gradient signals. In the Fokker-Planck formulation, this is modeled as an initial delta-function injection at the smallest context scale, reflecting that all initial predictive capability is concentrated at the shortest context scale. This is directly analogous to energy injection at large scales ($k\approx 0$) in turbulence (Appendix A) and to hierarchical mass assembly from small scales ($m_h\approx 0$) in cosmic structure formation (Appendix B).

As shown in Fig. 6, with increasing training, capability carriers migrate outward to larger r as the model learns to exploit longer-range dependencies (Phase I, upscale cascade), then compress back toward smaller r as large-scale capability is distilled into efficient local representations (Phase II, Mode A downscale cascade), while Mode B continues advancing carriers to larger r at the learning frontier. At each scale r, the model must wait a finite "time" τ_r until informative patterns at the next scale $r+dr$ appear in the data stream and can be productively learned—driving a jump of *capability carriers* to the next scale $r\to r+dr$. This nonlinear learning dynamics is therefore represented as a one-dimensional continuous-time random walk in r-space: a local jump $r\to r+dr$ corresponds to acquiring infinitesimal predictive capability at scale r, reducing the corresponding loss. In direct analogy with ordinary diffusion, where spatial random walks generate the evolution of a particle density, $E_X(r,s)$ describes the temporal evolution of the capability distribution in context-scale space.

This construction naturally incorporates three key properties:

(i) *Scale-locality.* Learning at scale r builds incrementally on the representation already consolidated at scale $r-dr$: capability transfers locally through adjacent context-scale shells, with strongly non-local jumps being sub-leading. In the random-walk picture, this is implemented by restricting transitions to $r\to r\pm dr$, so that long-range dependencies are assembled hierarchically from short-range ones rather than acquired directly.

(ii) *Scale-dependent waiting time.* The waiting time $\tau_r(r)$ between successive jumps is the characteristic learning time at scale r. We parameterize it as $\tau_r(r)\propto r^{-\gamma}$ with $\gamma>0$: shorter context scales are learned more slowly because fewer independent informative cues are available, while larger scales carry more independent information and exhibit shorter waiting times. This is the simplest closure admitting a self-similar learning process, in direct analogy with the turbulence eddy lifetime $\tau_k\propto k^{-\gamma}$ (Appendix A). The exponent γ is determined self-consistently in Section 3.4.

(iii) *Scale-independent cascade rate.* A hallmark of nonequilibrium systems at statistical steady state is a scale-independent cascade rate: in turbulence, the energy flux ε is constant across the inertial range (Appendix A); in cosmic structure formation, the mass flux ε_m is independent of halo mass scale m_h (Appendix B). By direct analogy, the loss cascade rate

$$\varepsilon_X = \frac{E_X\cdot r}{\tau_r} = \frac{\Delta L_r\cdot P_X\cdot r}{\tau_r} \tag{10}$$

be scale-independent, ensuring no capability accumulates at any intermediate scale—a necessary condition for self-similar learning. With $\tau_r\propto r^{-\gamma}$, scale-independence of ε_X immediately requires

$$P_X\propto r^{-(1+\gamma)}, \tag{11}$$

which is the inertial-range scaling of the loss spectrum, directly analogous to the Kolmogorov $-5/3$ energy spectrum in turbulence.

3.2 Autonomous Langevin and Fokker–Planck Equations

We model the evolution of the loss spectrum in context-scale space as a one-dimensional continuous-time random walk in which the walker's position r represents the effective context scale, with a scale-dependent waiting time τ_r between transitions. The corresponding Langevin equation in r-space reads

$$\frac{dr}{d\tau} = \sqrt{2D_L(r)}\,\varsigma(\tau),\quad \langle\varsigma(\tau)\varsigma(\tau')\rangle=\delta(\tau-\tau') \tag{12}$$

where τ is the effective training time (related to optimizer step s or model parameter N) and $\varsigma(\tau)$ is Gaussian white noise. The position-dependent diffusivity is $D_L(r)=D_0\,r^{2\gamma}$, arising from the assumption $\tau_r\propto r^{-\gamma}$: a smaller r yields a smaller diffusivity and hence a longer waiting time, reflecting the slower learning rate at short context scales. Equation (12) is Markovian in r. The waiting time depends only on the current scale r, not on the history of the walk, leading to the Fokker-Planck equation for the loss distribution.

In direct analogy with diffusion, where the Fokker-Planck equation governs the evolution of a probability density driven by Brownian motion, the loss distribution $P_X(r,\tau)$ satisfies a generalized Fokker-Planck equation with a position-dependent diffusivity that reflects the scale-dependent nature of the learning process,

$$\frac{\partial P_X(r,\tau)}{\partial\tau} = \frac{\partial}{\partial r}\left[\alpha\frac{\partial D_L}{\partial r}P_X + D_L\frac{\partial P_X}{\partial r}\right], \tag{13}$$

where α is a stochastic-calculus interpretation parameter ($\alpha=1$ for Itô, $\alpha=1/2$ for Stratonovich [23], with other values determined by the underlying microphysics).

3.3 Analytical Solutions for Loss Spectrum

3.3.1 Normalized spectrum and effective scale ξ

Combining the scale-independence requirement (Eq. (11)) with the general solution (Appendix C) yields exact solution of Eq. (13)

$$P_X(r,\tau) = \frac{A}{\xi}\left(\frac{r}{\xi}\right)^{-(1+\gamma)}\exp\left[-\left(\frac{r}{\xi}\right)^{2-2\gamma}\right], \tag{14}$$

where A is a normalization constant. This solution gives an inertial-range-like scaling $E_X(r,\tau)\propto P_X(r,\tau)\propto r^{-(1+\gamma)}$ at small r, controlled by the waiting-time exponent γ, together with an exponential cutoff $\exp(-r^{2-2\gamma})$ at large r. The time-dependent characteristic (effective) scale $\xi\equiv\xi(\tau)$ marks the onset of exponential decay in the loss spectrum, and evolves with time τ as

$$\xi(\tau) = \left[(2-2\gamma)^2 D_0\,\tau\right]^{\frac{1}{2-2\gamma}} \propto \tau^{\frac{1}{2-2\gamma}}. \tag{15}$$

A key structural property of the solution (Eq. (14)) is that time enters *exclusively* through the effective scale $\xi(\tau)$: the time dependence of $P_X(r,\tau)$ is encoded in this single evolving scale. Consequently, the direction of the cascade is entirely controlled by whether ξ grows or shrinks. This observation motivates a non-monotonic mapping between Langevin time τ and optimizer step s:

$$\tau(s) \propto \begin{cases} s^{+1} & \text{Phase I} \quad (s < s^*), \\ s^{-1} & \text{Phase II Mode A: Downscale} \quad (s > s^*), \\ s^{+1} & \text{Phase II Mode B: Upscale} \quad (s > s^*). \end{cases} \tag{16}$$

In Phase I, a single effective context scale $\xi(s)$ advances monotonically as the model progressively recruits longer-range dependencies. At the critical step s^*, ξ bifurcates into two branches: Mode B sustains the upscale advance with an expanding frontier scale $\xi_B \propto s^{+3/5}$, while Mode A reverses direction with a contracting compression scale $\xi_A \propto s^{-3/5}$. This bifurcation produces the non-monotonic evolution of the usable context scale for Mode A (Fig. 8): ξ first expands as the model discovers long-range structure (Phase I), then contracts with s as it compresses that structure into efficient short-range representations (Phase II).

The nth moment of loss distribution $P_X(r,\tau)$ reads

$$\int_r^\infty P_X(r',\tau)\, r'^n\, dr' = \frac{A\xi^n}{(2-2\gamma)}\,\Gamma\left(\frac{n-\gamma}{2-2\gamma}, \left(\frac{r}{\xi}\right)^{2-2\gamma}\right), \tag{17}$$

where $\Gamma(x,y)$ is the upper incomplete Gamma function with the limiting behaviors

$$\begin{aligned} &\lim_{y\to 0}\Gamma(x,y) = \Gamma(x) - y^x/x \qquad (x>0), \\ &\lim_{y\to 0}\Gamma(-x,y) = \big(y^{-x} - \Gamma(1-x)\big)/x \qquad (0<x<1), \end{aligned} \tag{18}$$

and $\Gamma(x, y\to\infty) = 0$.

Taking ξ_b as the infrared cutoff for the smallest r, the normalization constant A is obtained from $\int_{\xi_b}^\infty P(r,\tau)\,dr = 1$ (zeroth moment, $n=0$) using Eq. (17):

$$A = \frac{(2-2\gamma)}{\Gamma\left(-\frac{\gamma}{2-2\gamma}, \left(\frac{\xi_b}{\xi}\right)^{2-2\gamma}\right)} \approx \frac{\gamma}{\left(\frac{\xi_b}{\xi}\right)^{-\gamma} - \Gamma\left(\frac{2-3\gamma}{2-2\gamma}\right)}. \tag{19}$$

3.3.2 *Two cumulative loss functions: Λ_X and Λ_Y*

The scale-resolved training dynamics are characterized by two complementary cumulative loss functions, each answering a distinct question about the model's predictive performance.

(i) *Mode A: Context-dependent cumulative loss* $\Lambda_X(r,\tau)$. The first cumulative loss measures how much additional error the model incurs when its context is restricted to scale r, compared to using the full context window:

$$\begin{aligned} \Lambda_X(r,\tau) &= L_r(r,\tau) - L_\infty(\tau) = \Delta L_r(\tau) \int_r^\infty P_X(r',\tau)\, dr' \\ &= C_X(\tau)\cdot\Gamma\left(-\frac{\gamma}{2-2\gamma}, \left(\frac{r}{\xi_A}\right)^{2-2\gamma}\right), \end{aligned} \tag{20}$$

where $C_X(\tau) = \frac{A}{2-2\gamma}\Delta L_r(\tau)$ is a time-dependent prefactor and $\xi_A(\tau)$ is the effective context compression scale of Mode A.

Physically, $\Lambda_X(r,\tau)$ quantifies the **cost of context removal**: it measures the predictive penalty incurred by physically restricting the model's receptive field to the most recent r tokens. This penalty arises because information at scales $r' > r$ is no longer available to the model. Figure 5 plots Λ_X for two different models. The two fitting parameters $C_X(\tau)$ and $\xi_A(\tau)$ are extracted by fitting Eq. (20) to the scale-resolved Λ_X at each checkpoint (Fig. 8).

(ii) *Mode B: Residual full-context cumulative loss* $\Lambda_Y(r,\tau)$. The second cumulative loss decomposes the full-context loss floor $L_\infty(\tau)$ across context scales:

$$\begin{aligned} \Lambda_Y(r,\tau) &= \int_r^\infty E_Y(r',\tau)\, dr' = L_\infty(\tau)\int_r^\infty P_Y(r',\tau)\, dr' \\ &= C_Y(\tau)\cdot\Gamma\left(-\frac{\gamma}{2-2\gamma}, \left(\frac{r}{\xi_B}\right)^{2-2\gamma}\right), \end{aligned} \tag{21}$$

where $C_Y(\tau) = \frac{A}{2-2\gamma}L_\infty(\tau)$, and $\xi_B(\tau)$ is the frontier absorption scale of Mode B. The spectrum $E_Y(r,\tau)$ is the rate density at which residual full-context loss is distributed per unit context scale, and $P_Y(r,\tau) = E_Y(r,\tau)/L_\infty(\tau)$ is its normalized form.

Physically, $\Lambda_Y(r,\tau)$ quantifies the **cost of incomplete learning**: it measures how much of the model's residual full-context loss $L_\infty(\tau)$ is attributable to its failure to perfectly exploit dependencies at scales $r' > r$, *even though full context is available in the input*. The crucial distinction is: Λ_X measures the cost of *removing* context (context availability); Λ_Y measures the cost of not yet having *learned to use* available context (learned capacity).

(iii) *Scale structure of* $L_\infty(\tau)$. The full-context loss floor $L_\infty(s)$ reflects the model's residual failure to exploit statistical dependencies across all context scales simultaneously, despite having full context available. This residual loss has internal scale structure, resolved by $\Lambda_Y(r,\tau)$. As training progresses, the frontier $\xi_B(s)$ advances upward and the model progressively masters longer-range dependencies, reducing $L_\infty(s)$.

(iv) *Shared functional form.* $P_X(r,\tau)$ and $P_Y(r,\tau)$ satisfy the same Fokker-Planck solution (Eq. (13)), sharing the same waiting-time exponent and the same inertial-range spectrum $E \propto r^{-(1+\gamma)}$ with exponential cutoff (Eq. (14)); they differ only in their effective scale and amplitude, and evolve in opposite temporal directions:

	Mode A (Λ_X)	Mode B (Λ_Y)
Effective scale	ξ_A (contracting)	ξ_B (expanding)
Amplitude	ΔL_r	L_∞
Process	Internal compression	Frontier absorption

In what follows, we present analytical solutions for Mode A; solutions for Mode B follow from the replacements $\xi_A \to \xi_B$ and $\Delta L_r \to L_\infty$.

3.3.3 *Analytical Solutions for Loss Flux*

By direct analogy with the constant-flux inertial range of turbulence, we define the *spectral loss flux* $\Pi_X(r,\tau)$ as the net rate of loss transfer across scale r—from all scales $r' < r$ to all scales $r' > r$—which, in a statistically steady cascade, is conserved across the inertial range:

$$\begin{aligned} \Pi_X(r,\tau) = -\frac{\partial\Lambda_X}{\partial\tau} = -\frac{A}{2-2\gamma}\Bigg[&\frac{\partial\Delta L_r}{\partial\tau}\,\Gamma\left(-\frac{\gamma}{2-2\gamma}, \left(\frac{r}{\xi_A}\right)^{2-2\gamma}\right) \\ &+ (2-2\gamma)\,\frac{\Delta L_r}{\tau}\,\frac{\partial\ln\xi_A}{\partial\ln\tau}\left(\frac{r}{\xi_A}\right)^{-\gamma}\exp\left(-\left(\frac{r}{\xi_A}\right)^{2-2\gamma}\right)\Bigg]. \end{aligned} \tag{22}$$

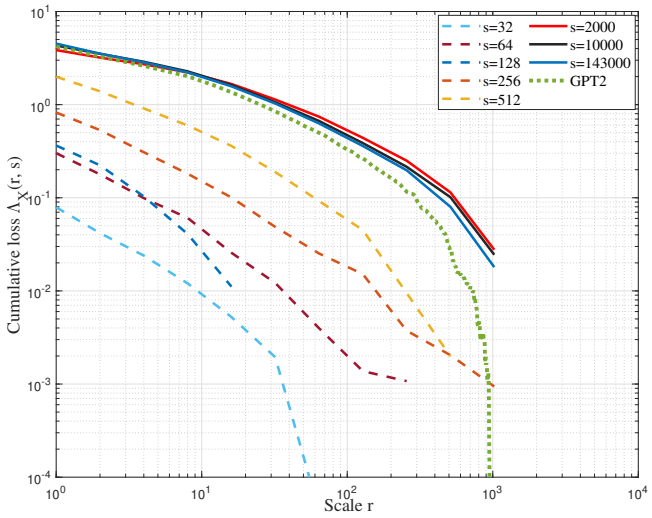


Figure 5. Context-dependent cumulative loss $\Lambda_X(r,s) = L_r(r,s) - L_\infty(s)$ versus context scale r at selected training steps s for Pythia-160M. **Phase I** ($s < s^*$, dashed lines): Λ_X grows monotonically across all scales, reflecting an upscale cascade in which the model progressively extends its predictive reach to larger context distances. **Phase II** ($s > s^*$, solid lines): Λ_X at large scales decreases while short-scale structure is retained, consistent with Mode A downscale compression in which learned long-range structure is consolidated into efficient local representations. The dotted green line shows the converged GPT-2 model for reference. Fitting Eq. (20) to Λ_X yields the Mode A parameters $\xi_A(s)$ and $C_X(s)$, whose evolution is shown in Fig. 8.

This can be re-expressed compactly in terms of Λ_X and P_X:

$$\Pi_X(r,\tau) = -\frac{\Delta L_r}{\tau}\frac{\partial \ln \xi_A}{\partial \ln \tau}\left[P_X(r,\tau)\cdot r + \frac{\Lambda_X}{\Delta L_r}\cdot\frac{\partial \ln \Delta L_r}{\partial \ln \xi_A}\right]. \tag{23}$$

The spatial derivative of the loss flux is

$$\frac{\partial \Pi_X}{\partial r} = \frac{\Delta L_r}{\tau}\frac{\partial \ln \xi_A}{\partial \ln \tau}P_X(r,\tau)\left[\frac{\partial \ln \Delta L_r}{\partial \ln \xi_A} - 1 - \frac{\partial \ln P_X}{\partial \ln r}\right]. \tag{24}$$

The scale r_m at which Π_X is maximized satisfies

$$\left.\frac{\partial \Pi_X}{\partial r}\right|_{r_m} = 0, \qquad \frac{\partial \ln \Delta L_r}{\partial \ln \xi_A} = -\gamma - (2-2\gamma)\left(\frac{r_m}{\xi_A}\right)^{2-2\gamma}. \tag{25}$$

A hallmark of the statistically steady cascade state in Phase II is a scale-independent loss flux: $\partial \Pi_X/\partial r = 0$ for all r in the inertial range. Imposing this condition on Eq. (24) yields a fundamental constraint relating ΔL_r to the effective scale ξ_A:

$$\frac{\partial \Pi_X}{\partial r} \equiv 0 \quad \Rightarrow \quad \frac{\partial \ln \Delta L_r}{\partial \ln \xi_A} = 1 + \frac{\partial \ln P_X}{\partial \ln r} = -\gamma. \tag{26}$$

This is a central result: it uniquely fixes the Phase II scaling $\Delta L_r \propto \xi_A^{-\gamma}$ for Mode A and $L_\infty \propto \xi_B^{-\gamma}$ for Mode B as a direct consequence of the scale-independent cascade, without any additional assumptions.

3.3.4 Scale-Independent Loss Flux and Training Dynamics

To see this clearly, in the small-scale limit, the loss flux (Eq. (22)) simplifies to

$$\varepsilon_X(\tau) \equiv \Pi_X(r\to 0,\tau) \approx \underbrace{\frac{A}{\gamma}\Gamma\left(\frac{2-3\gamma}{2-2\gamma}\right)\frac{\partial \Delta L_r}{\partial \tau}}_{(1)} \underbrace{- \frac{A}{\gamma}\left(\frac{r}{\xi_A}\right)^{-\gamma}\frac{\partial \Delta L_r}{\partial \tau}\left(1 + \frac{\gamma}{\frac{\partial \ln \Delta L_r}{\partial \ln \xi_A}}\right)}_{(2)}. \tag{27}$$

Term (1) is scale-independent and positive (since ΔL_r increases with time), corresponding to a *downscale (direct) cascade* from large to small r, analogous to the energy cascade in three-dimensional turbulence (Appendix A). Term (2) is scale-dependent, negative, and grows in magnitude as $r \to 0$, corresponding to an *upscale (inverse) cascade* from small to large r, analogous to hierarchical structure formation in cosmology (Appendix B). The competition between these two terms drives the two phases of training. To illustrate, a normalized loss flux is defined as

$$\bar{\varepsilon}_X(\tau) = \frac{\varepsilon_X}{\frac{A}{\gamma}\cdot\frac{\partial \Delta L_r}{\partial \tau}} = \Gamma\left(\frac{2-3\gamma}{2-2\gamma}\right) - \left(\frac{r}{\xi_A}\right)^{-\gamma}\left(1 + \frac{\gamma}{\frac{\partial \ln \Delta L_r}{\partial \ln \xi_A}}\right). \tag{28}$$

The evolution of ε_X, ΔL_r, and ξ during training is illustrated in Figs. 6 and 7, and proceeds in two stages:

(i) **Phase I: Initial (upscale) cascade.** Term (2) dominates, giving $\varepsilon_X < 0$ with magnitude decreasing toward larger r (Fig. 7). The model learns rapidly from training data: ΔL_r rises steeply and the effective scale ξ expands quickly. The model broadens its learning across scales. Since $\varepsilon_X \approx 0$ near $r \sim \xi$, we can use Eq. (27) evaluated at $r = \xi$ to derive the loss scaling:

$$\frac{\partial \ln \Delta L_r}{\partial \ln \xi} = \beta = \frac{\gamma}{\Gamma\left(\frac{2-3\gamma}{2-2\gamma}\right) - 1}. \tag{29}$$

With $\xi(\tau) \propto \tau^{1/(2-2\gamma)}$ (Eq. (15)) and $\tau \propto s$ in Phase I,

$$\Delta L_r = B\cdot\xi^\beta \propto \tau^{\frac{\beta}{2-2\gamma}} \propto s^{\frac{\beta}{2-2\gamma}}, \tag{30}$$

where $B \equiv B(N)$ is a model-capacity prefactor: larger models achieve greater context gain ΔL_r at any given frontier scale ξ because more parameters can encode multi-scale dependencies. $B(N)$ increases with N but saturates as $N \to \infty$. (N1: Eq. (29) uses the small-r form Eq. (27) at $r = \xi$, where it is no longer accurate: with the exact flux Eq. (22), $\Pi_X(\xi) \propto -[\Gamma(-0.1,1) + (2-2\gamma)e^{-1}/\beta]$ with $\Gamma(-0.1,1) = 0.210$ (the small-r form gives -0.686), which is negative for every $\beta > 0$. The value $\beta \approx 2.43$ is therefore an estimate from the leading-order flux, not an exact zero-flux condition.)

(ii) **Bifurcation of training dynamics after s^*.** After the critical step s^*, the training dynamics in Phase II bifurcates into two simultaneous processes governed by the same Fokker-Planck structure (same γ) but operating on different components of the loss with opposite time orientations:

Phase II (Mode A): Downscale cascade (Refinement). Term (1) dominates, giving a scale-independent $\varepsilon_X > 0$ (Fig. 7): the cascade is directed from large to small r. The system reaches a statistically steady state with a scale-independent cascade rate $\varepsilon_X > 0$, which decreases over time. The scale-independence of ε_X imposes

$$\frac{\partial \ln \Delta L_r}{\partial \ln \xi_A} = -\gamma, \qquad \text{equivalently,} \quad \Delta L_r \propto \xi_A^{-\gamma}. \tag{31}$$

Since the maximum possible loss is set by the vocabulary size, this constraint takes the form

$$\Delta L_r = \ln(N_{\text{voc}})\,\xi_A^{-\gamma}. \tag{32}$$

This scaling is expected to be universal: independent of model size and dataset size, depending only on the multiscale structure of the training data through γ. All models with sufficient representation capabilities trained on the same data will converge to the same scaling exponent. With $\tau \propto s^{-1}$ in Phase II,

$$\Delta L_r = \ln(N_{\text{voc}})\,\xi_A^{-\gamma} \propto \tau^{-\frac{\gamma}{2-2\gamma}} \propto s^{\frac{\gamma}{2-2\gamma}}. \tag{33}$$

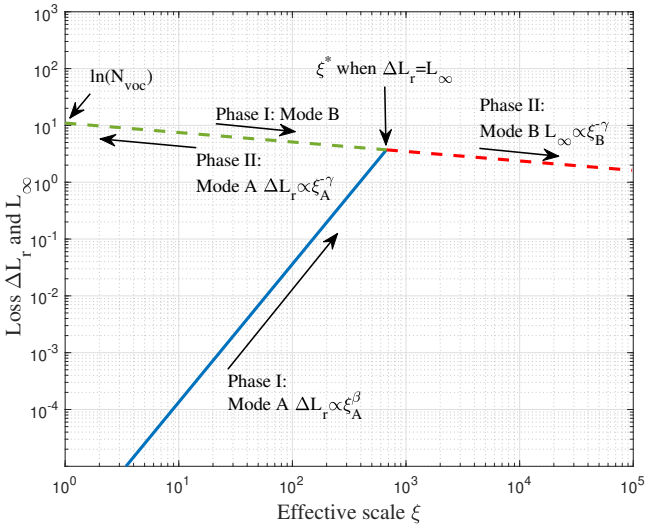


Figure 6. Context-dependent loss ΔL_r and full-context loss L_∞ versus effective scale ξ across training. **Phase I** ($s < s^*$, $\Delta L_r < L_\infty$): the two modes are coupled, with $\xi_A = \xi_B = \xi$ growing together. $\Delta L_r \propto \xi_A^\beta$ ($\beta \approx 2.43$ for $\gamma = 1/6$) rises steeply as the model rapidly builds its loss spectrum across scales (Mode A, depth), while L_∞ falls as the learning frontier expands (Mode B, breadth). **Phase II** ($s > s^*$, $\Delta L_r > L_\infty$): training reaches a statistically steady state and bifurcates into two decoupled modes running in opposite directions, analogous to the turbulent inertial range. Mode A: ξ_A contracts while $\Delta L_r \propto \xi_A^{-\gamma}$ grows slowly, as the model compresses large-scale structure into local representations. Mode B: ξ_B expands while $L_\infty \propto \xi_B^{-\gamma}$ falls slowly, as the frontier keeps absorbing new long-range structure. The phases meet at the critical scale ξ^* (the maximum of ξ_A), given by Eq. (36).

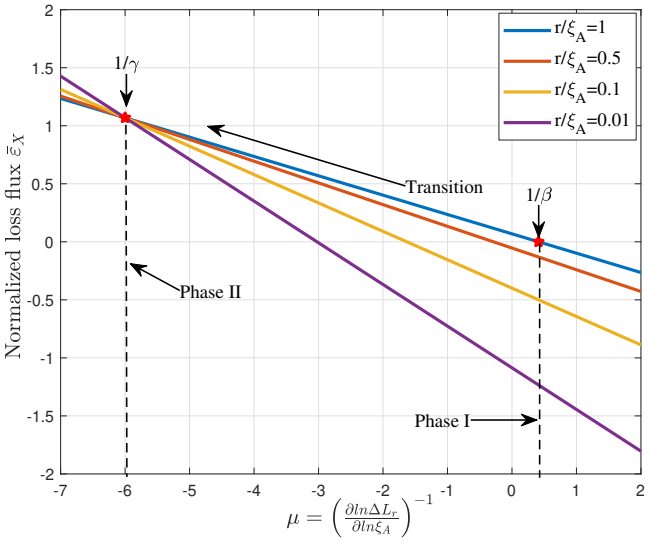


Figure 7. Normalized loss flux $\bar{\varepsilon}_X$ as a function of the inverse scaling exponent $\mu = (\partial \ln \Delta L_r / \partial \ln \xi_A)^{-1}$, plotted for four representative scale ratios $r/\xi_A \in \{0.01, 0.1, 0.5, 1.0\}$. Using $\gamma = 1/6$ as an example, the two training phases are clearly delineated. **Phase I** (right region, $\mu = 1/\beta$): $\bar{\varepsilon}_X < 0$ across all scales $r < \xi_A$, with magnitude increasing toward smaller r/ξ_A, reflecting a scale-dependent upscale cascade in which information propagates from short to long context distances. **Phase II** (left region, $\mu = -1/\gamma$): $\bar{\varepsilon}_X$ converges to a positive, scale-independent value for all r/ξ_A, reflecting a statistically steady downscale cascade in which loss is transferred from large to small context scales at a uniform rate. The transition between the two phases coincides with the critical values s^* or ξ^*, where the cascade direction reverses, and the effective scale ξ_A reaches its maximum value ξ^*.

The effective scale ξ_A decreases with s, while ΔL_r slowly increases: the model compresses diffuse large-scale representations into efficient short-range ones, deepening local understanding at the cost of dissolving redundant long-range structure. This refinement phase can, in principle, extend indefinitely, exhibiting clear power-law scaling, unless bottlenecked by model capacity or compute.

Phase II (Mode B): Frontier absorption (upscale cascade). Simultaneously with Mode A, the model continues to absorb new predictive structure from the data stream at the expanding frontier scale ξ_B. This mode is a continuation of the Phase I upscale dynamics, characterized by a negative (upscale) scale-independent cascade rate (similarly to Eq. (27) for ε_X for Mode A),

$$\varepsilon_Y(\tau) \equiv \Pi_Y(r \to 0, \tau)$$
$$\approx \underbrace{\frac{A}{\gamma}\Gamma\left(\frac{2-3\gamma}{2-2\gamma}\right)\frac{\partial L_\infty}{\partial \tau}}_{(1)} \underbrace{- \frac{A}{\gamma}\left(\frac{r}{\xi_B}\right)^{-\gamma}\frac{\partial L_\infty}{\partial \tau}\left(1 + \frac{\gamma}{\frac{\partial \ln L_\infty}{\partial \ln \xi_B}}\right)}_{(2)}. \tag{34}$$

When term (1) dominates, the scale-independent rate satisfies $\varepsilon_Y < 0$, indicating an inverse cascade that transfers information toward larger scales. The frontier scale grows as $\xi_B(\tau) \propto \tau^{1/(2-2\gamma)}$ (Eq. (15)), and the scale-independent cascade condition requires a vanishing term (2) that gives (similar to Eq. (32))

$$L_\infty = \ln(N_{\text{voc}})\,\xi_B^{-\gamma} \propto \tau^{-\frac{\gamma}{2-2\gamma}} \propto s^{-\frac{\gamma}{2-2\gamma}}. \tag{35}$$

As training progresses, ξ_B expands outward while L_∞ decreases: the model progressively masters longer-range dependencies by absorbing new long-range predictive structure from the data. This power-law decrease extends indefinitely in principle until L_∞ approaches the entropy rate of natural text—the irreducible loss of an ideal generative process on the true data distribution. For self-determined value of $\gamma = 1/6$ (Eq. (45)), Eq. (35) provides the theoretical basis for the empirical Kaplan-style scaling law $L \propto D^{-0.095}$ observed in LLMs.

(iii) **Critical transition.** Equating the loss scalings from Phases I and II at the transition, $\ln(N_{\text{voc}})\,\xi_A^{-\gamma} = B\,\xi_A^{\beta}$, gives

$$\xi^* = \left(\frac{\ln N_{\text{voc}}}{B}\right)^{\frac{1}{\beta+\gamma}}, \qquad \Delta L_r^* = B^{\frac{\gamma}{\beta+\gamma}}\left[\ln N_{\text{voc}}\right]^{\frac{\beta}{\beta+\gamma}}. \tag{36}$$

Larger models (larger B) reach a smaller critical scale ξ^*. For the Pythia-160M model, with $\gamma = 1/6$, $\beta \approx 2.43$, and $B \approx 5 \times 10^{-7}$, we obtain $\xi^* \approx 668$ and $\Delta L_r^* \approx 3.66$. The normalization constant at the transition (with $\xi_b = 1$) is

$$A = \frac{(2-2\gamma)}{\Gamma\left(-\frac{\gamma}{2-2\gamma}, \left(\frac{1}{\xi^*}\right)^{2-2\gamma}\right)}, \tag{37}$$

yielding $A \approx 0.1$ for Pythia-160M (Eq. (19)).

This two-phase dynamics bears a qualitative resemblance to human learning: Phase I mirrors rapid, broad acquisition of knowledge across all context scales, while Phase II mirrors its slower, depth-first consolidation into efficient local representations alongside a continued, slower expansion of the frontier. In Phase II, the model reaches a statistically steady cascade state — learned representations become statistically homogeneous across context scales, governed by a scale-independent mean cascade rate — in direct analogy with the

inertial range of turbulence, where energy cascades at a constant rate ε between the injection and dissipation scales.

3.4 Waiting Time τ_r and Self-Consistent Value $\gamma = 1/6$

The waiting time τ_r at context scale r is the characteristic time for the model to make one unit of progress in extracting and encoding statistical dependencies at context distance r. We decompose it as

$$\tau_r = \frac{1}{\Gamma_r \cdot P_r}, \tag{38}$$

where Γ_r is the *attempt frequency*, the rate at which scale-r information in the training stream attempts to drive the model's current internal state toward a learning update at scale r. While P_r is the *success probability* per attempt, i.e., the probability that a single attempt produces a genuine learning update at scale r. Each attempt takes on average $1/\Gamma_r$, and the number of attempts needed for the first success is geometrically distributed with mean $1/P_r$; combining the two gives the mean waiting time τ_r.

(i) *Attempt frequency.* The attempt frequency Γ_r quantifies the rate that is driven by scale-r information contained in the model's current state. In direct analogy with the eddy velocity $u_k \propto \sqrt{E(k)k}$ in turbulence, which sets the rate of energy transfer across scales (Appendix A), we define the characteristic *velocity* at scale r as

$$u_r = \sqrt{E_X(r) \cdot r}, \tag{39}$$

where $E_X(r) \cdot r$ measures the residual loss per logarithmic scale interval at r, equivalently, the density of unresolved predictive uncertainty per decade of context scale. A larger $E_X(r)r$ means more unresolved predictive structure remains at scale r. Therefore, a larger u_r implies a greater mismatch between the model's current predictions and the targets at scale r, producing stronger gradient signals per encounter and hence a higher rate of attempted learning updates. The attempt frequency is therefore proportional to this characteristic velocity,

$$\Gamma_r \propto u_r = \sqrt{E_X(r) \cdot r}, \tag{40}$$

where $E_X(r) = \Delta L_r \cdot P_X(r)$.

(ii) *Success probability and Heaps' law.* The success probability P_r measures the probability that a single scale-r attempt produces a genuine learning update. It is controlled by the amount of novel or not-yet-reliably-learned predictive information supplied by the data at scale r: the more independent cues remain unlearned, the larger the prediction-target mismatch, and hence the stronger the gradient signal and larger probability to overcome the local learning barrier.

Natural language has heavy-tailed, approximately Zipfian token statistics [22], and token-token mutual information decays with separation distance. These facts imply diminishing returns from increasing context length. We therefore model the effective number ($M(r)$) of statistically independent informative cues in a context window of length r by the scaling ansatz

$$M(r) \propto r^{\alpha_H}, \qquad \alpha_H \approx 0.4\text{–}0.6. \tag{41}$$

Here α_H is identified with the empirical Heaps exponent for vocabulary growth [24], which is itself closely related to Zipfian word-frequency statistics. Strictly, Heaps' law counts distinct word types rather than independent predictive cues; thus Eq. (41) should be interpreted as a leading-order modeling ansatz. The justification is that both vocabulary growth and the emergence of new predictive cues are governed by the same heavy-tailed linguistic statistics, while the decay of mutual information with distance limits the independent information contributed by each additional token.

Different long-range linguistic patterns (coreference, topic coherence, syntactic agreement) correspond to distinct directions in the model's high-dimensional parameter space and therefore contribute approximately independent (orthogonal) gradient signals ($\|\mathbf{G}_r\|$). Since these $M(r)$ contributions are approximately orthogonal, they add incoherently via RMS aggregation,

$$\|\mathbf{G}_r\| \propto \sqrt{M(r)} \propto r^{\alpha_H/2}, \tag{42}$$

rather than coherently, because the diversity of linguistic structures at scale r ensures their gradient directions are not systematically aligned. The success probability is therefore proportional to

$$P_r \propto \|\mathbf{G}_r\| \propto r^{\alpha_H/2}. \tag{43}$$

Thus, the waiting time τ_r is determined by two complementary factors: the attempt frequency Γ_r, which reflects how strongly the model's current internal state is driven by scale-r information, and the success probability P_r, which reflects how much independent predictive information the data supplies at that scale. Together they determine τ_r through Eq. (38).

(iii) *Self-consistent value* $\gamma = 1/6$. Substituting the attempt frequency Eqs. (40) and success probability (43) into Eq. (38):

$$\tau_r = \frac{1}{\Gamma_r \cdot P_r} \propto \frac{1}{\sqrt{E_X(r) \cdot r} \cdot r^{\alpha_H/2}} \propto r^{-\gamma}. \tag{44}$$

With $E_X \propto r^{-(1+\gamma)}$ (Eq. (11)),

$$\frac{\gamma - \alpha_H}{2} = -\gamma \quad \Rightarrow \quad \gamma = \frac{\alpha_H}{3} \approx \frac{1}{6}. \tag{45}$$

The waiting time $\tau_r \propto r^{-\gamma}$ thus emerges self-consistently from the heavy-tailed statistical structure of natural language (Heaps' law with $\alpha_H \approx 1/2$ combined with RMS gradient aggregation), yielding the universal exponent $\gamma = \alpha_H/3 = 1/6$. The loss spectrum $P_X \propto E_X \propto r^{-(1+\gamma)} \propto r^{-7/6}$ can be confirmed by the data in Fig. 4.

4 PREDICTING THE SCALING LAWS

4.1 Scaling laws for mode A and mode B.

With $\gamma = \alpha_H/3 = 1/6$ determined self-consistently, parameter $\beta \approx 2.43$ (Eq. (29)). From Eqs. (30) and (33), scalings laws read

$$\begin{aligned}
&\Delta L_r \propto \xi^{\beta} \propto s^{\frac{\beta}{2-2\gamma}} \propto s^{1.46} \quad \text{(Phase I)},\\
&\Delta L_r \propto \xi_A^{-\gamma} \propto s^{\frac{\gamma}{2-2\gamma}} \propto s^{\frac{\alpha_H}{6-2\alpha_H}} \propto s^{1/10} \quad \text{(Phase II Mode A)},\\
&L_\infty \propto \xi_B^{-\gamma} \propto s^{-\frac{\gamma}{2-2\gamma}} \propto s^{-\frac{\alpha_H}{6-2\alpha_H}} \propto s^{-1/10} \quad \text{(Phase II Mode B)}.
\end{aligned} \tag{46}$$

Recall that $\Delta L_r(s)$ measures the total predictive gain from exploiting multi-scale context beyond a single token—the depth of the model's context dependence. It grows rapidly in Phase I as the model expands its predictive reach across scales, then continues to grow slowly as $\Delta L_r \propto s^{1/10}$ in Phase II as learned representations are compressed and consolidated (Fig. 3). Meanwhile, $L_\infty(s)$ measures the model's residual inability to exploit statistical dependencies across all context scales despite full context being available—the breadth of its frontier exploration—decreasing as $L_\infty \propto s^{-1/10}$ in Phase II as the model progressively masters longer-range dependencies (Fig. 3). Together, ΔL_r and L_∞ characterize two complementary dimensions

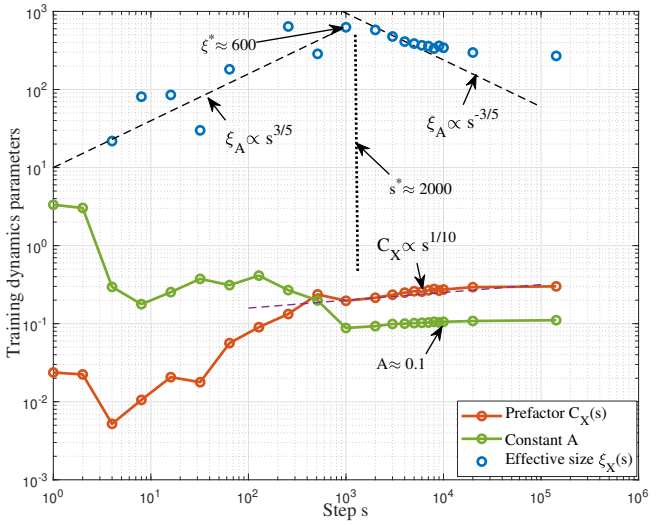


Figure 8. Evolution of the effective context scale $\xi_A(s)$ and cumulative-loss prefactor $C_X(s)$ with training step s, extracted from fitting Eq. (20) to the Pythia-160M validation losses. **Phase I** ($s < s^*$): $\xi_A(s)$ grows rapidly, reflecting the fast expansion of the model's predictive reach to large context distances (upscale cascade). The prefactor $C_X(s)$ increases rapidly, indicating rapid accumulation of total model capacity. **Phase II** ($s > s^*$): $\xi_A(s)$ contracts while $C_X(s)$ grows slowly (fitted slopes in Section 4; the dashed lines are the predicted $s^{-3/5}$ and $s^{1/10}$). This phase corresponds to depth-first refinement and downscale compression of the learned information spectrum, in agreement with theoretical predictions. Dashed lines indicate the predicted power-law scalings in each phase.

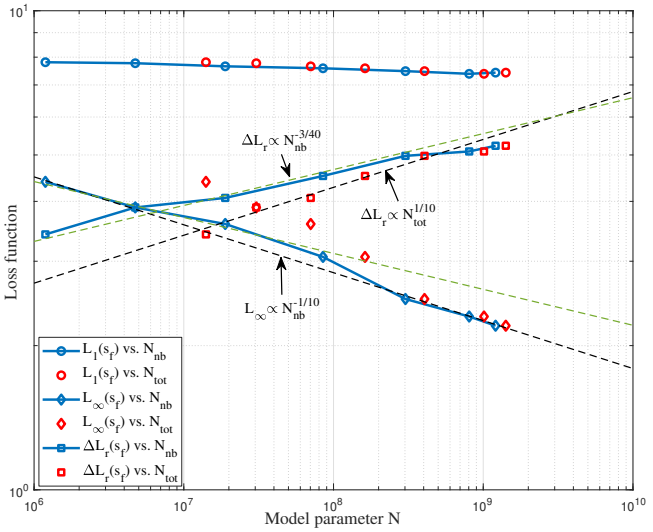


Figure 9. Loss components L_1, L_∞, and ΔL_r versus model size for Pythia models (14M-1.4B parameters) at convergence ($s_f \approx 143{,}000$ steps, ∼300B tokens). Red and blue symbols denote total (N_{tot}) and non-embedding (N_{nb}) parameter counts. Green dashed lines plots $\pm\frac{3}{40}$ scaling, and black dashed lines plots $\pm\frac{1}{10}$ scaling. L_1 saturates across all model sizes, showing that even the smallest models capture local token statistics. L_∞ decreases monotonically as $N_{\text{nb}}^{-1/10}$, consistent with $\alpha_N = 1/10$ and a continuously expanding learning frontier, while ΔL_r grows as $N_{\text{tot}}^{+1/10}$ or $N_{\text{nb}}^{+3/40}$, reflecting increasing multi-scale context utilization. The exponent $1/10$ ($\lambda = 1$) implies each parameter contributes equally to the Langevin time, whereas $3/40$ ($\lambda = 3/4$) points to sublinear capacity gains from non-embedding parameters specifically; larger model suites are needed to resolve this distinction conclusively.

of learning: *depth* (how effectively the model uses the context it has) and *breadth* (how far its learning frontier has advanced).

N2: exponents measured from the plotted data. Pythia-160M (Fig. 3): $s^* \approx 1.9 \times 10^3$ (between the checkpoints 1000 and 2000); Phase I, $\Delta L_r \propto s^{0.90}$ for $s = 16$–1000; Phase II, $\Delta L_r \propto s^{+0.075}$ and $L_\infty \propto s^{-0.101}$ for $s = 2000$–10^4, flattening to $s^{+0.036}$ and $s^{-0.046}$ over $s = 2000$–1.43×10^5. Loss spectrum (Fig. 4, $r = 2$–32): slopes -1.05, -1.12, -1.18 at $s = 2000$, 10^4, 1.43×10^5 and -1.10, -1.21, -1.19 for converged Pythia-70M, Pythia-410M and GPT-2, against $-7/6$; at larger r the cutoff steepens the spectrum. The training curves of the other Pythia models (14M–1.4B, author's loss tables) give Phase II slopes ($s = 2000$–10^4) of L_∞ from -0.07 to -0.16 and of ΔL_r from -0.06 to $+0.09$, so the exponent $1/10$ is not yet seen to be independent of model size.

Fitting the two-parameter cumulative loss function $\Lambda_X(r,s) = L_r(r,s) - L_\infty(s)$ (Eq. (20)) to held-out losses from the Pythia-160M validation set yields the evolution of the effective scale $\xi_A(s)$ and prefactor $C_X(s)$ with training step s, shown in Fig. 8 (N3: the plotted values are least-squares fits of Λ_X in linear scale over $r = 1$–1024 with the exponent $2 - 2\gamma$ fixed at 1.67, reproduced here to 10^{-4}; with these ξ_A, Eq. (32) gives $\ln(N_{\text{voc}})\xi_A^{-\gamma} = 0.93$–$0.98$ of the measured ΔL_r for $s \geqslant 2000$. Fitting $\ln \Lambda_X$ instead, which weights all scales equally, gives $\xi_A = 780$–903 and an almost constant ξ_A (slope -0.02) for $s \geqslant 2000$, so the size of the contraction depends on the weighting of the small Λ_X at large r). In Phase I, both parameters grow rapidly, reflecting fast, broad acquisition of predictive capacity across all context scales. After the transition at s^*, $\xi_A(s)$ contracts while $C_X(s)$ continues to grow, but more slowly: the fitted slopes are -0.32 (ξ_A) and $+0.15$ (C_X) for $s = 2000$–10^4, flattening to -0.16 and $+0.07$ over $s = 2000$–1.43×10^5. The direction agrees with the Phase II picture; the contraction of ξ_A is shallower than the predicted $s^{-3/5}$ (dashed line in Fig. 8).

4.2 Connection to Kaplan-style scaling.

The empirical Kaplan-style scaling laws report the full-context held-out loss $L \equiv L_\infty(s)$ as a power law in training data D [1]. Our theory directly predicts this scaling via Mode B frontier absorption (Eq. (35)): in Phase II, the full-context loss decreases as

$$L_\infty(s) \propto s^{-\frac{\gamma}{2-2\gamma}}, \tag{47}$$

where $\gamma = \alpha_H/3$ is determined self-consistently from the Heaps exponent $\alpha_H \approx 1/2$ (Section 3.4). With $D \propto s$ (a natural identification for a fixed batch size per optimizer step, as used in the Pythia training schedule evaluated here; datasets trained with a varying batch size would require replacing s with cumulative tokens seen), the predicted loss-vs-data exponent is

$$\alpha_D = \frac{\gamma}{2-2\gamma} = \frac{\alpha_H}{6-2\alpha_H} = \frac{1}{10}, \tag{48}$$

in good agreement with the empirically reported value $L \propto D^{-0.095}$ [1, 3]. This agreement is non-trivial: the exponent $\alpha_D = 1/10$ is not fitted but derived entirely from the linguistic statistics of the training data, providing a mechanistic interpretation of scalings.

4.3 Incorporating Model Size N

The framework above treats optimizer step s as the time variable τ driving Mode B frontier expansion at fixed N. We now consider an analogous scenario obtained by exchanging the roles of N and D:

fix D sufficiently large that data is never the bottleneck, and instead treat model size N as an alternative "clock" driving the same Fokker-Planck dynamics. Increasing N at fixed D progressively enlarges the model's representational capacity, advancing the learning frontier ξ_B to larger context scales, just as increasing s does at fixed N: in both cases, a previously binding constraint is relaxed–insufficient data exposure in the s-scenario, insufficient representational capacity in the N-scenario. This is a structural analogy, not a claim that N and s are physically interchangeable. What the two scenarios share is the kinematic constant-flux constraint that fixes γ (Section 3.4); below tests whether this shared constraint produces structurally similar scaling laws in both cases, not that N and s are the same process.

4.3.1 *Universality of the Fokker–Planck structure.*

Both scenarios are governed by the same Fokker-Planck dynamics, because the cascade dynamics, including scale-local loss transfer, scale-dependent waiting times $\tau_r \propto r^{-\gamma}$, and inertial-range loss spectrum $E_X \propto r^{-(1+\gamma)}$, is determined entirely by the statistical structure of the training data (Zipf/Heaps laws), not by whether the frontier is advanced by data or parameters. The scale-independent rate of cascade condition at statistically steady state is a *kinematic* constraint: regardless of which resource is limiting, loss flux ε_X must be scale-independent across the inertial range (Eq. (10)).

4.3.2 *Loss-vs-N scaling.*

Applying Mode B scaling with $\tau \propto N^{\lambda}$ gives

$$L_\infty = \ln(N_{\rm voc})\,\xi_B^{-\gamma}, \quad \xi_B \propto N^{\lambda/(2-2\gamma)}, \tag{49}$$

and therefore

$$L_\infty \propto N^{-\alpha_N}, \quad \alpha_N = \frac{\lambda\gamma}{2-2\gamma}, \tag{50}$$

where $\lambda \leqslant 1$ characterizes the efficiency with which additional parameters advance the effective Langevin time: $\lambda = 1$ corresponds to every parameter contributing independently to representational capacity (the symmetric case $\alpha_N = \alpha_D$), while $\lambda = 3/4$ reflects sublinear capacity gains from parameter redundancy.

Figure 9 tests this prediction against Pythia models (14M-1.4B parameters) at convergence ($s_f \approx 143{,}000$ steps, ~300B tokens), fitting Eq. (49) separately to $L_\infty(N)$ and $\Delta L_r(N)$, and separately against total ($N_{\rm tot}$) and non-embedding ($N_{\rm nb}$) parameter counts. For L_∞, the fitted exponent against $N_{\rm nb}$ is -0.10, i.e. $\lambda \approx 1$ (against $N_{\rm tot}$ it is steeper, -0.155) — of the same order as, though somewhat larger than, the empirical Kaplan value $\alpha_N \approx 0.076$ [1]. A numerical match instead appears in ΔL_r, whose exponent depends on which count is used: $\Delta L_r \propto N_{\rm tot}^{+1/10}$ ($\lambda = 1$) versus $\Delta L_r \propto N_{\rm nb}^{+3/40}$ ($\lambda = 3/4$). Larger model suites spanning a wider range of N are needed to resolve it conclusively. (N4: least-squares slopes of the seven plotted points of Fig. 9: ΔL_r, $+0.090$ against $N_{\rm tot}$ and $+0.060$ against $N_{\rm nb}$; L_1, -0.012 and -0.008. The $\pm 1/10$ and $\pm 3/40$ lines of the figure are guides, not fits.)

The sublinear-redundancy interpretation of $\lambda = 3/4$ is accordingly a hypothesis rather than an established result. Under this interpretation, as N grows, a fraction of parameters reinforce existing representations rather than opening genuinely new ones, so the effective Langevin time scales as $N^{3/4}$ rather than N. This mirrors Kleiber's metabolic scaling law [25], where biological support networks grow sublinearly relative to active metabolic tissue. The analogy is suggestive rather than a formal derivation, and requires further work.

4.3.3 *Combined scaling law.*

When both N and D are finite, the effective Langevin time is set by whichever bottleneck is more restrictive. We treat $\tau_D \propto D$ and $\tau_N \propto N^{\lambda}$ as two independent, additive rate-limiting resources and combine them harmonically,

$$\frac{1}{\tau_{\rm eff}} = \frac{1}{\tau_D} + \frac{1}{\tau_N} = \frac{D_c}{D} + \left(\frac{N_c}{N}\right)^{\lambda}, \quad \lambda = \frac{\alpha_N}{\alpha_D}. \tag{51}$$

This is the simplest ansatz consistent with the known Chinchilla/Kaplan additive form; it is not itself derived from the Fokker–Planck equation. Together with $L \propto \tau_{\rm eff}^{-\alpha_D}$, it yields the scaling

$$L(N,D) = \left[\left(\frac{N_c}{N}\right)^{\lambda} + \frac{D_c}{D}\right]^{\alpha_D}. \tag{52}$$

Its additive structure reflects the harmonic combination of the two independent bottlenecks, while the outer exponent α_D is inherited from the Fokker–Planck relation $L \propto \tau^{-\alpha_D}$.

5 CONCLUSION

Motivated by the analogy between language model training and nonequilibrium statistical systems, particularly the energy cascade in turbulence and hierarchical structure formation in cosmology, we have developed a stochastic multiscale theory of transformer training dynamics. The central idea is that effective context utilization evolves as a biased random walk in context-distance scale space, governed by a Fokker-Planck equation whose scale-dependent diffusivity is inherited from the heavy-tailed (Zipfian) statistics of natural language. Decomposing the held-out loss into a context-dependent component $\Delta L_r(s)$ and a full-context component $L_\infty(s)$ yields a loss spectrum $E_X(r,s)$, in direct analogy with the turbulence energy spectrum, and shows how power-law loss scaling emerges from a scale-dependent waiting time $\tau_r \propto r^{-\gamma}$—the language-modeling counterpart of the eddy turnover time. The principal results are:

(i) **Two-phase cascade structure.** At a critical step s^*, where $\Delta L_r(s) = L_\infty(s)$, training reaches a statistically steady state and bifurcates into two simultaneous modes: Mode A (downscale cascade, compressing learned structure into local representations, $\xi_A \propto s^{-3/5}$) and Mode B (upscale frontier absorption, extending to longer-range dependencies, $\xi_B \propto s^{+3/5}$).
(ii) **Self-consistent scaling exponent.** The waiting-time exponent $\gamma = \alpha_H/3 = 1/6$ follows self-consistently from Heaps' law ($\alpha_H \approx 1/2$) and RMS gradient aggregation, fixing the loss spectrum $E_X(r,s) \propto r^{-7/6}$—the language-modeling analogue of the stochastic argument that fixes turbulence's $-5/3$ spectral exponent.
(iii) **Loss-vs-data scaling.** The theory predicts the Phase II exponent $\alpha_D = \gamma/(2-2\gamma) = 1/10$, giving $\Delta L_r(s) \propto s^{1/10}$ and $L_\infty(s) \propto s^{-1/10}$, in good agreement with the empirical Kaplan result $L \propto D^{-0.095}$.
(iv) **Scaling form from first principles.** The additive structure of $L(N,D) = [(N_c/N)^{\alpha_N/\alpha_D} + D_c/D]^{\alpha_D}$ follows from the harmonic combination of two independent bottlenecks—finite data and finite model capacity—acting on a single dynamical variable, the effective Langevin time $\tau_{\rm eff}$.
(v) **Analytically tractable spectrum and flux.** The cumulative loss, loss spectrum, and loss flux across scale all admit closed-form solutions, validated against Pythia checkpoints across training.

Beyond explaining existing scaling laws, this nonequilibrium perspective offers new diagnostics for long-context generalization by

separating depth of context dependence (ΔL_r) from breadth of frontier exploration (L_∞), and provides a principled basis for interventions in data selection, training curricula, and architecture design aimed at improving both learning modes jointly.

Major assumptions. The theory rests on several modeling choices rather than as derived necessities. First, the mapping from optimizer step s (and from model size N) onto a single Langevin time τ, including its non-monotonic form across phases, is a construction chosen to match observed behavior, not a result independently derived from the Fokker-Planck equation. Second, the self-consistent value $\gamma = 1/6$ rests on auxiliary assumptions: that Heaps' law is a valid proxy for the number of independent predictive cues at a given context scale, that gradient contributions from distinct long-range patterns aggregate incoherently (RMS/orthogonal), and that the attempt frequency Γ_r is proportional to an eddy-velocity-like quantity. Third, treating N as an alternative Langevin clock and combining the N- and D-limited regimes harmonically are structural analogies motivated by consistency with known empirical scaling forms, not first-principles derivations. Finally, the detailed within-training validation is currently restricted to a single model family (Pythia-160M for the dynamical fits, the wider Pythia suite for the N-scaling fits); generalization across architectures remains to be established.

Future work. Priorities include: testing the universality of $\gamma = 1/6$ and the two-phase bifurcation across architecturally distinct model families rather than a single family; deriving the $\tau(s)$ and $\tau(N)$ mappings, and the harmonic (N, D) combination, from an explicit joint model of data exposure and parameter capacity; testing the sublinear $N^{3/4}$ capacity-scaling hypothesis on larger, more finely spaced model suites to disambiguate it from the linear (N^1) case; and extending the framework beyond held-out cross-entropy to downstream task performance, to assess whether the depth/breadth decomposition predicts long-context generalization in practice.

AUTHOR CONTRIBUTIONS

Z.J. Xu performed the conception, data analysis, and writing.

FUNDING

This research was supported by the Laboratory Directed Research and Development at Pacific Northwest National Laboratory (PNNL). PNNL is a multiprogram national laboratory operated for the U.S. Department of Energy (DOE) by Battelle Memorial Institute under contract no. DE-AC05-76RL01830.

DATA AVAILABILITY

All data generated or analysed during this study are included in this article. Key architectural properties of selected language models used for scale-dependent cross-entropy analysis are presented in table 1.

APPENDIX A: ENERGY SPECTRUM IN TURBULENCE

Kolmogorov's $-5/3$ scaling follows from the assumption that small-scale turbulence is statistically homogeneous and isotropic, characterized solely by a scale-independent mean energy dissipation rate ε and kinematic viscosity ν. Here we show that the same result and the full spectral shape emerge from a random-walk formulation of the energy cascade, without invoking ε as an independent assumption.

The derivations of Appendices A and B are given in full, with comparisons to simulations, in Xu [20]. We model the turbulent energy cascade as a continuous-time random walk of fictitious "energy carriers" in wavenumber k-space. Each carrier transports an infinitesimal energy dE, and a step $k \to k + \delta k$ represents a local triadic energy transfer at scale k. The distribution $P(k,t)$ of carriers is proportional to the energy spectrum, or $P(k,t) = 2E(k,t)/u_0^2$, where u_0^2 is the large-scale characteristic velocity. Four physical properties are built into this construction:

(i) *Scale-locality.* Energy transfer is dominated by local triadic interactions ($|\mathbf{k}| \sim |\mathbf{p}| \sim |\mathbf{q}|$), so carriers jump only between adjacent wavenumber shells.
(ii) *Energy conservation.* Triadic interactions conserve energy; the total number of carriers is conserved, and the dynamics merely redistributes energy across scales.
(iii) *Viscous dissipation.* Viscosity introduces a scale-dependent sink $2\nu k^2 P(k)$, reflecting the Laplacian structure of the Navier–Stokes equations: dissipation is negligible at large scales and dominant at small scales.
(iv) *Scale-dependent waiting time.* The waiting time between successive steps is identified with the eddy turnover time at scale k, parametrized as $\tau_{gk}(k) \propto k^{-\gamma}$ ($\gamma > 0$), encoding the faster decorrelation of small-scale eddies. The exponent γ is determined self-consistently below.

The derivation can be formulated in the same theoretical framework for nonequilibrium systems as we develop the nonequilibrium training dynamics for the loss spectrum of the language model.

(i) **Langevin and Fokker-Planck Equations.** With waiting time $\tau_{gk}(k) \propto k^{-\gamma}$, the k-space diffusivity is $D_P(k) = D_0 k^{2\gamma}$, and the Langevin equation for the carrier position reads

$$\frac{dk_t}{dt} = \sqrt{2D_P(k_t)}\,\xi(t), \qquad D_P(k) = D_0 k^{2\gamma}, \tag{A1}$$

where $\xi(t)$ is Gaussian white noise. The corresponding Fokker–Planck equation governing $P(k,t)$ is

$$\frac{\partial P}{\partial t} = \frac{\partial}{\partial k}\left[\alpha \frac{\partial D_P}{\partial k} P + D_P \frac{\partial P}{\partial k}\right] - 2\nu k^2 P, \tag{A2}$$

where α specifies the stochastic calculus. The steady-state solution of Eq. (A2) reads (Appendix C)

$$P(k) = A k^{-(1+\gamma)} \exp\left[-\lambda \left(\frac{k}{k_\eta}\right)^{2-\gamma}\right], \tag{A3}$$

where A is a normalization constant, $\lambda = \sqrt{2\nu/D_0}\, k_\eta^{2-\gamma}/(2-\gamma)$, and k_η marks the onset of exponential decay. The energy spectrum naturally follows the power-law scaling $E(k) \propto k^{-(1+\gamma)}$ for small k, with slope determined by γ. Crucially, the solution uniquely fixes parameter $\alpha = (1 + 1/\gamma)/2$.

(ii) **Self-Consistent Determination of $\gamma = 2/3$.** For a spectrum $E(k) \propto k^{-(1+\gamma)}$, the eddy velocity scales as $u_k \propto \sqrt{E(k)k} \propto k^{-\gamma/2}$, giving an eddy turnover time $\tau_k \propto (u_k k)^{-1} \propto k^{\gamma/2-1}$. Self-consistency requires the waiting time $\tau_{gk} \propto k^{-\gamma}$ to equal τ_k:

$$\frac{\gamma}{2} - 1 = -\gamma \quad \Longrightarrow \quad \gamma = \tfrac{2}{3}. \tag{A4}$$

This is the unique fixed point at which the Langevin equation is consistent with the nonlinear transfer rate generated by its

Table 1. Key architectural properties of selected language models used for scale-dependent cross-entropy analysis. d_{model}: hidden dimension; d_{ff}: feed-forward (intermediate) dimension; n_{heads}: number of query attention heads (numbers in parentheses give the number of key/value heads when grouped-query attention (GQA) is used, i.e. $n_{\text{heads}} \neq n_{\text{kv}}$); d_{head}: per-head dimension; n_{layers}: number of transformer layers; $r_{\max}$: maximum context length (tokens); V: vocabulary size; N_{total}: total parameter count (embeddings included), matching each model's official release name; $N_{\text{non-emb}}$: parameter count excluding embedding (and, where untied, unembedding) matrices. All models use a decoder-only (causal) transformer architecture.

Model	d_{model}	d_{ff}	n_{heads}	d_{head}	n_{layers}	$r_{\max}$	V	N_{tot}	N_{nb}	Pos. enc.	Activation	Trained on
GPT-2 Small	768	3072	12	64	12	1024	50,257	124M	85M	Learned abs.	GELU	WebText
Pythia-70M	512	2048	8	64	6	2048	50,304	70M	18.9M	RoPE	GELU	The Pile
Pythia-160M	768	3072	12	64	12	2048	50,304	160M	85.1M	RoPE	GELU	The Pile
Pythia-410M	1024	4096	16	64	24	2048	50,304	410M	302.3M	RoPE	GELU	The Pile
SmolLM2-135M	576	1536	9 (3)	64	30	8192	49,152	135M	106.7M	RoPE	SwiGLU	FineWeb-Edu, DCLM, The Stack

own stationary spectrum. It also naturally yields the Kolmogorov spectrum $E(k) \propto k^{-5/3}$.

(iii) **Energy Flux and Dissipation Rate.** In the inertial range $k_L \ll k \ll k_\eta$, the spectral energy flux (from Eq. (A3))

$$\begin{aligned}\Pi_E(k) &= \int_k^\infty 2\nu k'^2 E(k')\,dk' \\ &= \frac{A\nu u_0^2 k_\eta^{2-\gamma}}{(2-\gamma)\lambda}\exp\left[-\lambda\left(\frac{k}{k_\eta}\right)^{2-\gamma}\right]\end{aligned} \tag{A5}$$

is independent of k for $k \ll k_\eta$, i.e. throughout the inertial range, and falls off only near the dissipation scale, $\Pi_E(k)/\varepsilon = \exp[-\lambda(k/k_\eta)^{2-\gamma}]$. This is a consequence of the steady-state Fokker-Planck equation, not an assumption. The mean dissipation rate $\varepsilon = \Pi_E(k \to 0)$ emerges as

$$\varepsilon = \Pi_E(k \to 0) = \frac{A\nu u_0^2}{(2-\gamma)\lambda} k_\eta^{2-\gamma}, \tag{A6}$$

which is scale-independent in the inertial range, and a derived quantity rather than an input.

(iv) **Full Energy Spectrum.** Setting $\gamma = 2/3$ in Eq. (A3) and expressing the result in terms of ε and η yields

$$E(k) = C_K\,\varepsilon^{2/3}k^{-5/3}\exp\left[-\tfrac{3}{2}C_K(\eta k)^{4/3}\right], \tag{A7}$$

which is precisely the Pao spectrum [15], recovered analytically without any additional assumptions. Thus, a random walk of energy carriers driven by local triadic interactions reproduces the complete equilibrium energy spectrum, providing a natural framework for nonequilibrium turbulence.

APPENDIX B: MASS FUNCTION FOR DM HALOES

Hierarchical cosmic structure formation can be viewed as a random walk in halo mass space: a dark matter halo of mass m_h grows by accreting a single dark matter particle of mass m, stepping to mass $m_h + m$, with an average waiting time $\tau_g(m_h)$ between successive accretion events. This generates a continuous flux of mass toward larger scales. The resulting halo mass distribution is governed by a Fokker-Planck equation, in direct analogy with the turbulence energy spectrum of Sec. A and the loss spectrum of training dynamics.

(i) **Langevin and Fokker–Planck Equations.** We take the scale factor a as the natural time variable, so that the Langevin equation for halo mass reads

$$\frac{\partial m_h}{\partial a} = \sqrt{2D_p(m_h)}\,\varsigma(a) \propto \frac{m}{\tau_g(m_h)}, \tag{B1}$$

where $\varsigma(a)$ is Gaussian white noise. We assume a power-law waiting time $\tau_g(m_h) \propto m_h^{-\lambda}$ ($\lambda > 0$), reflecting the faster accretion of more massive haloes, which implies a mass-dependent diffusivity

$$D_p(m_h) = D_{p0}\,m_h^{2\lambda}. \tag{B2}$$

The corresponding Fokker–Planck equation for the halo mass function $f_M(m_h, a)$ is

$$\frac{\partial f_M}{\partial a} = \frac{\partial}{\partial m_h}\left[\alpha\frac{\partial D_p}{\partial m_h}f_M + D_p\frac{\partial f_M}{\partial m_h}\right], \tag{B3}$$

where α specifies the stochastic interpretation. Equation (B3) admits the analytic steady-state solution (Appendix C)

$$f_M(m_h, a) = \frac{m_h^{-\lambda}}{\sqrt{\pi D_{p0}\,a}}\exp\left[-\frac{m_h^{2-2\lambda}}{4(1-\lambda)^2 D_{p0}\,a}\right], \tag{B4}$$

which requires $\alpha = 1/2$ (Stratonovich interpretation). Introducing the characteristic mass (similar to the effective scale ξ for LLM)

$$m_h^* = \left[\frac{(1-\lambda)^2 D_{p0}\,a}{\eta_0}\right]^{1/(2-2\lambda)}, \tag{B5}$$

the mass function takes the compact form

$$f_M(m_h, a) = \frac{1-\lambda}{\sqrt{\pi\eta_0}\,m_h^*}\left(\frac{m_h}{m_h^*}\right)^{-\lambda}\exp\left[-\frac{1}{4\eta_0}\left(\frac{m_h}{m_h^*}\right)^{2-2\lambda}\right], \tag{B6}$$

with a power-law low-mass slope $\propto m_h^{-\lambda}$ and an exponential high-mass cutoff, both controlled by the single exponent λ.

(ii) **Self-Consistent Determination of λ.** The exponent λ is set by whichever physics dominates at the scale of interest. The effect of the primordial spectrum could be dominant on small scales. Here we consider a gravity-dominated regime for large haloes with $m_h \sim m_h^*$. Near the characteristic mass m_h^*, repeated mergers and smooth accretion erase memory of the initial conditions. The dominant physics is then set by virial equilibrium and large-scale bulk motions. The virial theorem gives $\sigma_v^2(m_h) \propto Gm_h/r_{\text{vir}} \propto m_h^{2/3}$, while the inter-halo bulk velocity dispersion σ_h^2 is relatively insensitive to m_h. The transition scale m_h^* is defined by velocity-dispersion matching:

$$\sigma_v^2(m_h^*, a) = \sigma_h^2(m_h^*, a), \tag{B7}$$

which yields the peak-height variable

$$\nu \simeq \frac{\sigma_v^2(m_h)}{\sigma_h^2} \simeq \left(\frac{m_h}{m_h^*}\right)^{2/3}, \tag{B8}$$

and hence

$$2 - 2\lambda = \tfrac{2}{3} \quad\Longrightarrow\quad \lambda = \tfrac{2}{3}. \tag{B9}$$

This dynamical value corresponds to an effective quasi-universal spectrum $P(k) \propto k^{-1}$ ($n = -1$), directly observed as a plateau in

the compensated power spectrum $P(k)k$ in N-body simulations. For $\lambda = 2/3$, Eq. (B5) gives $m_h^*(a) \propto a^{3/2}$, and the waiting time

$$\tau_g(m_h, a) \propto \frac{r_h}{G m_h} \propto m_h^{-2/3} a, \tag{B10}$$

where $r_h \propto m_h^{1/3} a$ is the comoving halo radius. This encodes both the intrinsic gravitational dynamics (deeper potential wells accrete faster) and the dilution of merger rates by cosmic expansion.

(iii) **Full Halo Mass Function.** Since λ transitions from the primordial value $\lambda_1 = (3-n)/6$ at low masses to the dynamical value $\lambda_2 = 2/3$ near m_h^*, we introduce a double-λ generalization of Eq. (B6):

$$f_M(m_h) = \frac{2(1-\lambda_2)}{\Gamma(q/2)} \frac{(2\sqrt{\eta_0})^{-q}}{m_h^*} \left(\frac{m_h^*}{m_h}\right)^{\lambda_1} \exp\left[-\frac{1}{4\eta_0}\left(\frac{m_h}{m_h^*}\right)^{2-2\lambda_2}\right], \tag{B11}$$

where $q = (1-\lambda_1)/(1-\lambda_2)$. Setting $\nu = (m_h/m_h^*)^{2-2\lambda_2}$, this reduces to a one-sided gamma distribution

$$f_{D\lambda}(\nu) = \frac{(2\sqrt{\eta_0})^{-q}}{\Gamma(q/2)} \nu^{q/2-1} \exp\left(-\frac{\nu}{4\eta_0}\right), \tag{B12}$$

whose shape is fully determined by λ_1, λ_2, and η_0. In the PS limit ($\lambda_1 = \lambda_2 = (3-n)/6$, $\eta_0 = 1/2$), Eq. (B11) reduces exactly to the Press–Schechter form. The double-λ mass function therefore captures both the primordial-spectrum-dominated low-mass regime and the gravity-dominated transition regime near m_h^* within a single, unified Fokker–Planck framework.

APPENDIX C: EXACT SOLUTIONS OF GENERAL FOKKER-PLANCK EQUATION

We present exact solutions of a general d-dimensional Fokker-Planck equation in scale space x, with a power-law diffusivity $D_P(x) \propto x^{2\gamma}$ and a possible dissipation term. This general solution can be applied to different nonequilibrium systems following the same principles.The equation reads

$$\frac{\partial P}{\partial t} = \left(\frac{\partial}{\partial x} + \frac{d-1}{x}\right)\left[\alpha \frac{\partial D_P}{\partial x} P + D_P \frac{\partial P}{\partial x}\right] - 2\nu x^2 P(x,t), \tag{C1}$$
$$D_P(x) = D_0 x^{2\gamma} = D_0 x^g,$$

where d is the dimension, α is the stochastic interpretation parameter, ν is the viscous dissipation. The associated waiting time obeys $\tau_g(x) \propto x^{-\gamma}$ with $g = 2\gamma$. We look for a similarity solution of

$$P(x,t) = \frac{A}{(D_0 t)^b x^a} \exp\left[-\frac{x^c}{f(D_0 t)^h}\right], \tag{C2}$$

where constant A can be fixed by normalization. Substituting Eq. (C2) into Eq. (C1) yields the algebraic constraint

$$\begin{aligned} &-f^2 (D_0 t)^{2h+1} x^g (a-\alpha g)(a-g+2-d) \\ &+fc(D_0 t)^{h+1} x^{c+g}(c+d-2-2a+g+\alpha g) \\ &-bf^2 (D_0 t)^{2h} x^2 + hf(D_0 t)^h x^{c+2} \\ &-c^2 (D_0 t) x^{2c+g} + 2f^2 (\nu/D_0)(D_0 t)^{2h+1} x^4 = 0. \end{aligned} \tag{C3}$$

For three cases, the parameters must take the following forms:

(i) Stationary turbulence spectrum ($d = 1$, $h = b = 0$, and $c \neq 0$):

$$a = 1+\gamma, \quad c = 2-\gamma, \quad \alpha = \frac{1}{2}\left(1+\frac{1}{\gamma}\right), \quad f = (2-\gamma)\sqrt{\frac{D_0}{2\nu}}. \tag{C4}$$

To prove that $\alpha = (1+1/\gamma)/2$ is the *unique* admissible value, we examine the algebraic constraint Eq. (C3) for the stationary case. First, $2c+g = 4$ such that the last two terms in Eq. (C3) can be canceled. The coefficient of x^g must vanish identically. This gives two roots: either $a = \alpha g = 2\alpha\gamma$, or $a = g-1 = 2\gamma-1$. For the physical inertial-range scaling $P(k) \propto k^{-(1+\gamma)}$ we need $a = 1+\gamma$. Therefore, the first root must hold: $1+\gamma = 2\alpha\gamma$, giving $\alpha = (1+\gamma)/(2\gamma)$. For $\gamma = 2/3$ this evaluates to $\alpha = 5/4$. This is a mathematical necessity, not a free choice.

(ii) LLM loss spectrum ($d = 1$, $b \neq 0$, $h \neq 0$, $c \neq 0$, and $\nu = 0$):

$$\begin{aligned} &a = 1+\gamma, \quad c = 2-2\gamma, \quad h = 1, \quad f = (2-2\gamma)^2, \\ &\alpha = \frac{1}{2}\left(1+\frac{1}{\gamma}\right), \quad b = \frac{1-2\alpha\gamma}{2-2\gamma}. \end{aligned} \tag{C5}$$

The coefficient of x^g must vanish identically. This gives two roots: either $a = \alpha g = 2\alpha\gamma$, or $a = g-1 = 2\gamma-1$. For scaling $P_X(r) \propto r^{-(1+\gamma)}$ we need $a = 1+\gamma$. Therefore, only the first root holds: $1+\gamma = 2\alpha\gamma$, giving $\alpha = (1+\gamma)/(2\gamma)$. For $\gamma = 1/6$ this evaluates to $\alpha = 7/2$.

(iii) Halo mass function ($d = 1$, $\nu = 0$, and $\alpha = 1/2$):

$$a = 2\alpha\gamma, \quad c = 2-2\gamma, \quad f = 4(1-\gamma)^2, h = 1, b = \frac{1-2\alpha\gamma}{2(1-\gamma)}. \tag{C6}$$

The Stratonovich convention $\alpha = 1/2$ is selected here because halo mergers arise as the continuum limit of a correlated (smooth) process with finite merger timescale, for which the Wong–Zakai theorem guarantees convergence to the Stratonovich interpretation.

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